

Independent Replication Report: *Spin-orbital Texture in Topological Insulators*

Zhang, Liu & Zhang, arXiv:1211.0762 (2012)

REPLICATE-PROJECT / TEXTURES-100 — from-equations reimplementaion

Compiled from `zhang2012_result.json` (physics run 2026-07-18)

Abstract

We independently reimplement the effective kp model of Zhang, Liu & Zhang (arXiv:1211.0762) from its published equations (Eqs. 1–10), with **no access to author code**. From the topological-insulator surface Dirac Hamiltonian $H_0 = \hbar v_f(\sigma_x k_y - \sigma_y k_x)$ we construct the six-dimensional orbital $\otimes$ spin eigenstates $|\Phi_{\pm}\rangle$ (Eqs. 4–5), form the orbital-resolved spin-projection tensor $D_{i,\eta}^{\pm}$ (Eq. 6) around a circular Fermi contour, and test every analytic prediction of the paper as a machine-checkable assertion. The p_z helical texture, the p_x/p_y [sin, cos] textures (Eqs. 7–8), the 2θ (π -periodic) orbital-character modulation, and the P_{p_x} polarization (Eq. 10) all reproduce to **machine precision** ($\sim 10^{-16}$) in angular form, with correct signs/handedness. One overall constant prefactor in the total in-plane spin magnitude differs by an exact factor of 2 — a convention gap (atom-sum collapse), not a physics disagreement.

VERDICT: REPLICATED. Coverage $\approx 8/10$, Agreement $\approx 9/10$. Core physics (spin-momentum locking + coupled spin-orbital texture: p_z helical, p_x/p_y 2θ orbital texture, tangential-upper / radial-lower dominance, right-handed in-plane orbital spin for both cones, P_{p_x} sign flip across the Dirac point) reproduced from the paper’s own equations.

1 The paper’s claims (the comparison anchors)

This is an analytic kp paper: its ”results” are closed-form expressions, not a single headline number. The anchors we reproduce:

1. **p_z helical spin texture:** left-handed for the upper Dirac cone, right-handed for the lower cone (same as the total surface-state spin texture).
2. **p_x/p_y orbital spin textures** follow a $[\sin \theta, \cos \theta]$ form (Eqs. 7–8) — they do *not* wind like p_z .
3. **Radial/tangential coupling:** for the upper cone a left-handed (right-handed) spin is coupled to the radial (tangential) orbital texture; exactly opposite for the lower cone.
4. **Dominance & net handedness:** the tangential (radial) orbital texture is dominant for the upper (lower) cone, making the *in-plane* orbital spin right-handed for *both* cones.
5. **P_{p_x} polarization (Eq. 10)** changes sign across the Dirac point (tangential $\rightarrow$ radial transition) — the proposed spin-resolved, photon-polarized ARPES observable.

2 Governing equations reimplemented

Surface Dirac Hamiltonian (isotropic; *no* hexagonal-warping term — see §6):

$$H_0 = \hbar v_f (\sigma_x k_y - \sigma_y k_x). \quad (1)$$

Helical zeroth-order spinors, $\theta = \text{atan2}(k_y, k_x)$:

$$|\uparrow_\theta\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} +ie^{-i\theta} \\ 1 \end{pmatrix}, \quad |\downarrow_\theta\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -ie^{-i\theta} \\ 1 \end{pmatrix}. \quad (2)$$

First-order-in- k orbital-resolved eigenstates (real fit parameters u_0, v_0, u_1, v_1, w_1 ; p_r radial, p_t tangential):

$$|\Phi_+\rangle = (u_0 - v_1 k)|p_z\rangle|\uparrow_\theta\rangle - \frac{i}{\sqrt{2}}(v_0 - u_1 k - w_1 k)|p_r\rangle|\uparrow_\theta\rangle + \frac{1}{\sqrt{2}}(v_0 - u_1 k + w_1 k)|p_t\rangle|\downarrow_\theta\rangle, \quad (3)$$

$$|\Phi_-\rangle = (u_0 + v_1 k)|p_z\rangle|\downarrow_\theta\rangle + \frac{i}{\sqrt{2}}(v_0 + u_1 k + w_1 k)|p_r\rangle|\downarrow_\theta\rangle - \frac{1}{\sqrt{2}}(v_0 + u_1 k - w_1 k)|p_t\rangle|\uparrow_\theta\rangle. \quad (4)$$

Orbital-resolved spin-projection tensor (the central object):

$$D_{i,\eta}^\pm(k, \theta) = \langle \Phi_\pm | (|p_i\rangle\langle p_i| \otimes s_\eta) | \Phi_\pm \rangle, \quad i \in \{p_x, p_y, p_z\}, \quad \eta \in \{0, x, y, z\}. \quad (5)$$

Analytic predictions tested:

$$[D_{p_z,x}^\pm, D_{p_z,y}^\pm, D_{p_z,z}^\pm] = \pm(u_0 \mp v_1 k)^2 [\sin \theta, -\cos \theta, 0], \quad (6)$$

$$[D_{p_x,\cdot}^\pm] = \mp \frac{v_0^2}{2} [\sin \theta, \cos \theta, 0], \quad [D_{p_y,\cdot}^\pm] = \pm \frac{v_0^2}{2} [\sin \theta, \cos \theta, 0] \quad (\text{small } k), \quad (7)$$

$$D_{p_x,0}^\pm - D_{p_y,0}^\pm = \mp 2 \cos(2\theta)(v_0 \mp k u_1) k w_1, \quad (8)$$

$$P_{p_x}^\pm = \mp \frac{2(v_0 \mp E u_1)(E w_1)}{(v_0 \mp E u_1)^2 + E^2 w_1^2}. \quad (\text{Eq.10})$$

3 Methods

Purely analytic / model-Hamiltonian, so run locally. We build $|\Phi_\pm\rangle$ as explicit 6-vectors (basis index $2 \cdot \text{orb} + \text{spin}$), form the 6×6 projectors $|p_i\rangle\langle p_i| \otimes s_\eta$, and evaluate $D_{i,\eta}^\pm$ on a 24-point circular Fermi contour ($\theta \in [0, 2\pi)$). Each paper formula is asserted as $\max_\theta |\text{numeric} - \text{analytic}|$. Conventions: $\hbar v_f = 1$; the atom-sum over α is collapsed to one effective orbital (rescales overall magnitudes only, see §6); representative parameters $u_0=1.0$, $v_0=0.6$, $u_1=0.4$, $v_1=0.3$, $w_1=0.25$, $k=0.3$ (small- k tests at $k=10^{-4}$). Runtime < 1 s.

4 Results: this work vs. paper

5 Comparison, conventions and scope (honest gaps)

1. **No hexagonal warping — because the paper has none.** The TEXTURES-100 task framing mentioned “hexagonal warping,” but arXiv:1211.0762 contains *only* the isotropic Dirac cone (Eq. 1) plus orbital-resolved wavefunctions. The Fu C_{3v} k^3 warping term is a *different* paper (Fu, PRL 2009). We deliberately replicated the *actual* paper; no warping term was implemented because none exists in it. If the intent was the Fu-warping model, that is a separate replication target.

Quantity (paper anchor)	This work	Paper	Status
p_z spin texture, angular form	err 2.2×10^{-16}	$\pm(u_0 \mp v_1 k)^2 [\sin, -\cos, 0]$	EXACT
p_x spin texture (Eq. 7), small k	err 3.9×10^{-5}	$\mp \frac{v_0^2}{2} [\sin, \cos, 0]$	MATCH
p_y spin texture (Eq. 8), small k	err 3.9×10^{-5}	$\pm \frac{v_0^2}{2} [\sin, \cos, 0]$	MATCH
Orbital-char. diff. (2θ)	err 2.1×10^{-16}	$\mp 2 \cos 2\theta (v_0 \mp ku_1)kw_1$	EXACT
Total in-plane spin, angular form	$[-\sin, \cos]$, RH both cones	$[-\sin, \cos]$, RH both	EXACT (form)
Total in-plane spin, prefactor	$2(v_0 \mp ku_1)kw_1$	$4(v_0 \mp ku_1)kw_1$	CONV. GAP ($\times 2$)
P_{p_x} polarization (Eq. 10)	err 2.5×10^{-16}	Eq. 10 closed form	EXACT
P_{p_x} signs (upper−/lower+)	confirmed	tangential−/radial+	CONFIRMED
$\langle \Phi_+ \Phi_+ \rangle$ (norm sanity)	1.064	—	consistent

Table 1: Per-claim comparison. “err” is $\max_\theta$ absolute deviation from the paper’s analytic formula over the Fermi contour. Small- k residuals (3.9×10^{-5}) are the $O(k)$ truncation of the paper’s stated small- k limit, verified at $k = 10^{-4}$.

- Factor-2 prefactor (convention gap, not physics).** Our total in-plane spin magnitude prefactor is $2(v_0 \mp ku_1)kw_1$ vs the paper’s $4(\dots)$ — an exact factor of 2, constant across k and θ . Direction and handedness are identical. Origin: the paper’s atom/orbital sum over α (a cross-term) is collapsed to one effective orbital here, and a Pauli-vs-spin ($\frac{1}{2}$) convention. Magnitudes are convention-dependent under this collapse; only ratios, angular forms and signs are physical — and those match.
- Small- k limit.** Eqs. 7–8 are the paper’s stated small- k forms; verified in that limit (err $\sim 4 \times 10^{-5}$ at $k = 10^{-4}$).
- No ab-initio (VASP) cross-check.** We compared analytic effective-model predictions only, as appropriate for a $k \cdot p$ replication; the paper’s DFT texture figures were not recomputed.

6 Reproducibility

```
# interpreter (numpy 2.3.5 / scipy 1.17.0):
/home/stevens/comfyui-env/bin/python \
  /home/stevens/textures-100/corpus/textures-polar-zhang2012/zhang2012_replicate.py
# -> writes zhang2012_checks.json; matches report/evidence/zhang2012_result.json
```

Evidence: `report/evidence/zhang2012_result.json` (verdict + per-claim), `zhang2012_checks.json` (raw errors), `zhang2012_replicate.py` (solver). No LaTeX engine is required to read this source; it compiles off-host with any standard `pdflatex`.